

# Supplemental Material for “Exact Replica Mechanics and Born-Record Large Deviations in Monitored Quantum Circuits”

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This Supplemental Material collects the reference data behind Secs. II.F and IV and Appendix A of the main text: the exceptional sets  $E_L$  of the three-replica compressed operator for  $d = 2$  and  $d = 3$  with blocks, minimal-polynomial degrees and exact ranks (Sec. SI), the per-block certificate data at  $L = 12$  (Sec. SII), the endpoint and  $p_\times$  ranks (Sec. SIII), the complete list of certificate files and scripts with SHA-256 hashes and reproduction commands (Sec. SIV); two proofs deferred from the main text (Secs. SV and SVI); the Clifford record-count runs behind the main-text Sec. IV.F (Sec. SVII), and the v22 exact-closure suite of that section (the reconstructed gap\_utils library with its validation battery, the conjugation-design test, the exact  $\bar{Z}_3$  cells and ladder, the  $\beta = 2$  identification cross-check, the marginal  $q = 4$  log-correction fits, the tilted-variance profile family, and the symmetry-reduced  $n = 5$   $L = 8$  block route; Sec. SVIID). Every number is exact (integer, modular with the stated prime, or algebraic with a certified isolating interval); no floating-point computation enters. Notation follows the main text:  $\mathbf{B}_p$  is the integer-polynomial matrix of Eq. (A1),  $\Gamma = H \times \mathbb{Z}_m$  with  $H = (S_3 \times S_3) \rtimes \mathbb{Z}_2$ , blocks are labelled  $(\rho, k)$  by an irreducible representation  $\rho$  of  $H$  (dimension  $d_\rho$ ;  $\iota\pm$  is the transposition parity, Ind an induced representation of dimension 2 or 4) and a momentum  $k$  of  $\mathbb{Z}_m$  ( $k = \pm 1$  and  $k = \pm 2$  denote the rational block that is the sum of the two conjugate complex sectors).

## SI. EXCEPTIONAL SETS

At a point  $p^* \in E_L$  the drop of rank  $\mathbf{B}_{p^*}$  below the generic rank is  $\sum d_\rho \nu_\rho$ , where  $\nu_\rho$  is the nullity of the block at  $p^*$ ; the tables list, for each  $p^*$ , the block(s) whose invariant factor vanishes, the degree of the minimal polynomial  $f$  of  $p^*$  over  $\mathbb{Q}$ , the nullity, and the exact rank of  $\mathbf{B}_{p^*}$ . The numerical value of  $p^*$  is a truncation of the certified isolating interval (all intervals have width below  $10^{-15}$ ). For  $d = 2$  every interior zero is simple (certificate multiplicity equal to the nullity in all 11 block factors at  $L = 8$  and the one at  $L = 6$ ); for  $d = 3$  all are simple except the linear factor  $3p - 2$  ( $p_\times(3) = 2/3$ ) in the  $k = \pm 1$  blocks  $(\text{triv} \otimes \text{triv})^{\iota+}$  at  $L = 6$ ,  $(\text{std} \otimes \text{std})^{\iota+}$  at  $L = 6, 8$  and  $(\text{sgn} \otimes \text{sgn})^{\iota+}$  at  $L = 8$  (file `n3_simple_zero_check.log`).

TABLE S1.  $E_8$  for  $d = 2$  (generic rank 1202): the eleven algebraic points in  $(0, 1)$ . For  $L = 6$ ,  $d = 2$ , the single point is  $p^* = (9 - \sqrt{41})/4 = 0.649219\dots$ , block  $\text{Ind}(\text{sgn} \otimes \text{std})$ ,  $k = \pm 1$ , nullity 2, rank 208;  $E_4 = \emptyset$ .

| $p^*$    | block $\rho$                                 | $k$     | deg $f$ | nullity | rank |
|----------|----------------------------------------------|---------|---------|---------|------|
| 0.255038 | $\text{Ind}(\text{triv} \otimes \text{std})$ | 0       | 9       | 1       | 1198 |
| 0.270503 | $(\text{std} \otimes \text{std})^{\iota+}$   | 0       | 14      | 1       | 1198 |
| 0.381759 | $(\text{triv} \otimes \text{triv})^{\iota+}$ | 0       | 19      | 1       | 1201 |
| 0.404429 | $\text{Ind}(\text{triv} \otimes \text{std})$ | 0       | 8       | 1       | 1198 |
| 0.510263 | $(\text{std} \otimes \text{std})^{\iota-}$   | $\pm 1$ | 6       | 2       | 1194 |
| 0.548044 | $(\text{triv} \otimes \text{triv})^{\iota+}$ | $\pm 1$ | 11      | 2       | 1200 |
| 0.557093 | $\text{Ind}(\text{triv} \otimes \text{std})$ | $\pm 1$ | 12      | 2       | 1194 |
| 0.684029 | $\text{Ind}(\text{triv} \otimes \text{std})$ | 2       | 5       | 2       | 1194 |
| 0.775468 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | $\pm 1$ | 11      | 2       | 1200 |
| 0.862494 | $\text{Ind}(\text{triv} \otimes \text{std})$ | $\pm 1$ | 12      | 2       | 1194 |
| 0.960754 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 0       | 22      | 1       | 1201 |

TABLE S2. Exceptional points for  $d = 3$  at  $L = 4$  (generic rank 21) and  $L = 6$  (generic rank 216). At  $L = 6$  the rational point  $p_\times(3) = 2/3$  affects seven blocks simultaneously— $(\text{triv} \otimes \text{triv})^{\iota+}$ ,  $k = \pm 1$  (nullity 2);  $\text{Ind}(\text{triv} \otimes \text{sgn})$ ,  $k = 0$  (1) and  $k = \pm 1$  (2);  $(\text{std} \otimes \text{std})^{\iota+}$ ,  $k = 0$  (1) and  $k = \pm 1$  (4);  $\text{Ind}(\text{triv} \otimes \text{std})$ ,  $k = 0$  (1) and  $k = \pm 1$  (2)—with total drop 40 and rank 176.

| $L$ | $p^*$    | block $\rho$                                 | $k$     | deg $f$ | $\nu$ | drop | rank |
|-----|----------|----------------------------------------------|---------|---------|-------|------|------|
| 4   | 0.779812 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 0       | 5       | 1     | 1    | 20   |
| 6   | 0.393762 | $\text{Ind}(\text{sgn} \otimes \text{std})$  | $\pm 1$ | 4       | 2     | 8    | 208  |
| 6   | 0.617321 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 0       | 14      | 1     | 1    | 215  |
| 6   | 2/3      | seven blocks (caption)                       |         | 1       |       | 40   | 176  |
| 6   | 0.744876 | $(\text{triv} \otimes \text{triv})^{\iota+}$ | 0       | 16      | 1     | 1    | 215  |
| 6   | 0.775117 | $(\text{std} \otimes \text{std})^{\iota+}$   | 0       | 13      | 1     | 4    | 212  |
| 6   | 0.826750 | $(\text{std} \otimes \text{std})^{\iota+}$   | $\pm 1$ | 2       | 2     | 8    | 208  |

## SII. THE $L = 12$ CERTIFICATE, BLOCK BY BLOCK

Table S4 lists, for the momentum- $\pi$  sector  $k = 3$  of  $L = 12$  ( $m = 6$ ), the multiplicity  $m_{\rho,3}$ , the reflection-parity dimensions  $(m_+, m_-)$ , the symmetry-forced bound  $2 \min(m_+, m_-)$ , the numbers  $(c_+, c_-)$  of exactly verified kernel vectors on the two parity sides, the resulting upper bound  $2 \min(m_+ - c_+, m_- - c_-)$  valid for all  $p$  outside a finite set, and the modular lower bound at  $p = 1/2$ ; the two bounds coincide in every block. The sectors  $k \neq 3$  have

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TABLE S3.  $E_8$  for  $d = 3$  (generic rank 1202): 55 points. At  $p_\times(3) = 2/3$  seven blocks lose rank simultaneously— $(\text{sgn} \otimes \text{sgn})^{\iota+}$ ,  $k = \pm 1$  (nullity 2);  $\text{Ind}(\text{triv} \otimes \text{sgn})$ ,  $k = 0$  (1) and  $k = \pm 1$  (4);  $(\text{std} \otimes \text{std})^{\iota+}$ ,  $k = 0$  (1) and  $k = \pm 1$  (6);  $\text{Ind}(\text{sgn} \otimes \text{std})$ ,  $k = 0$  (1) and  $k = \pm 1$  (4)—for a total drop of 60 and rank 1142. For the two degree-46 and degree-50 factors of  $(\text{std} \otimes \text{std})^{\iota+}$  the nullity is pinned by  $1 \leq \nu \leq \text{mult}_f(\det)$  and the evenness of  $\nu$  in  $k = \pm 1$  blocks, as stated in Appendix A.3.

| $p^*$    | block $\rho, k$                                     | deg $f$ | $\nu$ | drop | rank | $p^*$    | block $\rho, k$                                     | deg $f$ | $\nu$ | drop | rank |
|----------|-----------------------------------------------------|---------|-------|------|------|----------|-----------------------------------------------------|---------|-------|------|------|
| 0.019707 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$      | 11      | 1     | 4    | 1198 | 0.698975 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$      | 14      | 1     | 4    | 1198 |
| 0.043971 | $(\text{triv} \otimes \text{triv})^{\iota+}, 0$     | 49      | 1     | 1    | 1201 | 0.699932 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 |
| 0.059680 | $(\text{triv} \otimes \text{triv})^{\iota+}, \pm 1$ | 25      | 2     | 2    | 1200 | 0.717603 | $\text{Ind}(\text{triv} \otimes \text{sgn}), 0$     | 6       | 1     | 2    | 1200 |
| 0.114953 | $(\text{std} \otimes \text{std})^{\iota-}, \pm 1$   | 10      | 2     | 8    | 1194 | 0.726117 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$       | 40      | 1     | 1    | 1201 |
| 0.116570 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$     | 23      | 1     | 4    | 1198 | 0.727191 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$       | 40      | 1     | 1    | 1201 |
| 0.157295 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 | 0.727894 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$     | 12      | 1     | 4    | 1198 |
| 0.280527 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$ | 26      | 2     | 8    | 1194 | 0.734664 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 |
| 0.327751 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 | 0.736197 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$      | 11      | 1     | 4    | 1198 |
| 0.387717 | $\text{Ind}(\text{triv} \otimes \text{std}), 2$     | 10      | 2     | 8    | 1194 | 0.739118 | $\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$  | 28      | 2     | 8    | 1194 |
| 0.406403 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$       | 40      | 1     | 1    | 1201 | 0.747220 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$ | 26      | 2     | 8    | 1194 |
| 0.412493 | $\text{Ind}(\text{triv} \otimes \text{std}), 2$     | 10      | 2     | 8    | 1194 | 0.750329 | $(\text{std} \otimes \text{std})^{\iota-}, \pm 1$   | 10      | 2     | 8    | 1194 |
| 0.449885 | $(\text{std} \otimes \text{std})^{\iota-}, 0$       | 7       | 1     | 4    | 1198 | 0.752473 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 50      | 1     | 4    | 1198 |
| 0.451387 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 7       | 1     | 4    | 1198 | 0.778267 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$       | 40      | 1     | 1    | 1201 |
| 0.482738 | $(\text{std} \otimes \text{std})^{\iota-}, 0$       | 4       | 1     | 4    | 1198 | 0.803763 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$       | 40      | 1     | 1    | 1201 |
| 0.488163 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, 0$       | 3       | 1     | 1    | 1201 | 0.818251 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$      | 14      | 1     | 4    | 1198 |
| 0.499209 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$ | 26      | 2     | 8    | 1194 | 0.819702 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 |
| 0.562734 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$     | 12      | 1     | 4    | 1198 | 0.826750 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 2       | 2     | 8    | 1194 |
| 0.604733 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$     | 23      | 1     | 4    | 1198 | 0.838695 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 7       | 1     | 4    | 1198 |
| 0.611851 | $\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$  | 28      | 2     | 8    | 1194 | 0.839191 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 50      | 1     | 4    | 1198 |
| 0.616454 | $(\text{triv} \otimes \text{triv})^{\iota+}, 2$     | 14      | 2     | 2    | 1200 | 0.851727 | $\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$  | 28      | 2     | 8    | 1194 |
| 0.628181 | $(\text{triv} \otimes \text{triv})^{\iota+}, \pm 1$ | 25      | 2     | 2    | 1200 | 0.882507 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 |
| 0.630557 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 50      | 1     | 4    | 1198 | 0.893682 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 50      | 1     | 4    | 1198 |
| 0.630590 | $(\text{triv} \otimes \text{triv})^{\iota+}, 0$     | 49      | 1     | 1    | 1201 | 0.895616 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 7       | 1     | 4    | 1198 |
| 0.651297 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 7       | 1     | 4    | 1198 | 0.942564 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$ | 26      | 2     | 8    | 1194 |
| 0.658235 | $(\text{std} \otimes \text{std})^{\iota+}, 0$       | 50      | 1     | 4    | 1198 | 0.951754 | $(\text{triv} \otimes \text{triv})^{\iota+}, 0$     | 49      | 1     | 1    | 1201 |
| 0.664027 | $(\text{std} \otimes \text{std})^{\iota+}, \pm 1$   | 46      | 2     | 8    | 1194 | 0.951943 | $(\text{triv} \otimes \text{triv})^{\iota+}, 0$     | 3       | 1     | 1    | 1201 |
| 0.694261 | $(\text{sgn} \otimes \text{sgn})^{\iota+}, \pm 1$   | 25      | 2     | 2    | 1200 | 0.985582 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$ | 26      | 2     | 8    | 1194 |

full multiplicity rank, 38906 in total, so the generic rank is  $38906 + 7104 = 46010 = r_3(12) - 70$ . The block data are identical for  $d = 2$  and  $d = 3$  (files `n3_L12_final.json`, `n3_L12_final_d3.json`).

TABLE S4.  $k = 3$  sector at  $L = 12$ . Weighted totals: forced 7174, upper bound 7104, lower bound 7104.

| $\rho$                                       | $d_\rho$ | $m_{\rho,3}$ | $(m_+, m_-)$ | forced | $(c_+, c_-)$ | upper | lower |
|----------------------------------------------|----------|--------------|--------------|--------|--------------|-------|-------|
| $(\text{triv} \otimes \text{triv})^{\iota+}$ | 1        | 162          | (85,77)      | 154    | (0,0)        | 154   | 154   |
| $(\text{triv} \otimes \text{triv})^{\iota-}$ | 1        | 55           | (19,36)      | 38     | (0,3)        | 38    | 38    |
| $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 1        | 170          | (99,71)      | 142    | (4,0)        | 142   | 142   |
| $(\text{sgn} \otimes \text{sgn})^{\iota-}$   | 1        | 61           | (30,31)      | 60     | (1,0)        | 58    | 58    |
| $\text{Ind}(\text{triv} \otimes \text{sgn})$ | 2        | 237          | (132,105)    | 210    | (4,1)        | 208   | 208   |
| $(\text{std} \otimes \text{std})^{\iota+}$   | 4        | 545          | (304,241)    | 482    | (5,0)        | 482   | 482   |
| $(\text{std} \otimes \text{std})^{\iota-}$   | 4        | 328          | (164,164)    | 328    | (1,4)        | 320   | 320   |
| $\text{Ind}(\text{triv} \otimes \text{std})$ | 4        | 411          | (210,201)    | 402    | (0,3)        | 396   | 396   |
| $\text{Ind}(\text{sgn} \otimes \text{std})$  | 4        | 423          | (234,189)    | 378    | (7,1)        | 376   | 376   |

Table S5 gives the modular ranks of every block  $(\rho, k)$  at  $L = 12$  for three values of  $p$ : a generic point ( $p = 1/2$ ,  $d = 2$  and  $d = 3$  give identical tables), the endpoint  $p = 0$  ( $d = 2$ ; identical for the second prime 1048573), and  $p_\times(3) = 2/3$  at  $d = 3$  (identical for the second prime 1048573). Blocks with  $k$  and  $-k$  have equal rank; only  $k = 0, 1, 2, 3$  are listed. Modular ranks are lower bounds for the rank over  $\mathbb{Q}$ ; at  $p = 1/2$  they meet the upper bound of Table S4.

### III. ENDPOINT AND $p_\times$ RANKS

### SIV. FILES, HASHES AND REPRODUCTION

All certificate files are plain JSON or text. Table S8 lists them (63 files) with their full SHA-256 hashes; the same list, in `sha256sum` format, is the ledger `certificate_sha256.txt`, so that `sha256sum`

TABLE S5. Block ranks at  $L = 12$ : multiplicity  $m_{\rho,k}$  and rank at  $p = 1/2$  (generic),  $p = 0$ , and  $p = p_{\times}(3) = 2/3$  ( $d = 3$ ). Weighted totals  $\sum d_{\rho}$  rank over all  $k$ : 46010, 45822, 44898.

| block $\rho$                                 | $d_{\rho}$ | $m_{\rho,k}$ |     |     |     | rank, $p = \frac{1}{2}$ |     |     |     | rank, $p = 0$ |     |     |     | rank, $p = \frac{2}{3}, d = 3$ |     |     |     |
|----------------------------------------------|------------|--------------|-----|-----|-----|-------------------------|-----|-----|-----|---------------|-----|-----|-----|--------------------------------|-----|-----|-----|
|                                              |            | $k=0$        | 1   | 2   | 3   | 0                       | 1   | 2   | 3   | 0             | 1   | 2   | 3   | 0                              | 1   | 2   | 3   |
| $(\text{triv} \otimes \text{triv})^{\iota+}$ | 1          | 190          | 160 | 183 | 162 | 190                     | 160 | 183 | 154 | 190           | 160 | 183 | 154 | 187                            | 158 | 180 | 154 |
| $(\text{triv} \otimes \text{triv})^{\iota-}$ | 1          | 64           | 55  | 63  | 55  | 64                      | 55  | 63  | 38  | 62            | 54  | 61  | 38  | 58                             | 51  | 57  | 38  |
| $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 1          | 182          | 166 | 177 | 170 | 182                     | 166 | 177 | 142 | 178           | 163 | 172 | 142 | 171                            | 158 | 165 | 142 |
| $(\text{sgn} \otimes \text{sgn})^{\iota-}$   | 1          | 58           | 60  | 58  | 61  | 58                      | 60  | 58  | 58  | 58            | 60  | 58  | 58  | 58                             | 60  | 58  | 58  |
| $\text{Ind}(\text{triv} \otimes \text{sgn})$ | 2          | 222          | 233 | 218 | 237 | 222                     | 233 | 218 | 208 | 222           | 233 | 218 | 208 | 209                            | 223 | 205 | 206 |
| $(\text{std} \otimes \text{std})^{\iota+}$   | 4          | 560          | 545 | 560 | 545 | 560                     | 545 | 560 | 482 | 556           | 542 | 555 | 482 | 533                            | 525 | 532 | 480 |
| $(\text{std} \otimes \text{std})^{\iota-}$   | 4          | 322          | 328 | 322 | 328 | 322                     | 328 | 322 | 320 | 322           | 328 | 322 | 320 | 322                            | 328 | 322 | 320 |
| $\text{Ind}(\text{triv} \otimes \text{std})$ | 4          | 426          | 411 | 426 | 411 | 426                     | 411 | 426 | 396 | 426           | 411 | 426 | 396 | 423                            | 409 | 423 | 396 |
| $\text{Ind}(\text{sgn} \otimes \text{std})$  | 4          | 414          | 423 | 414 | 423 | 414                     | 423 | 414 | 376 | 410           | 420 | 409 | 376 | 395                            | 409 | 395 | 374 |

TABLE S6. Exact ranks of  $B_p$  at the endpoints and at  $p_{\times}(d) = (d+1)/[3(d-1)]$ . Generic ranks are certified for  $d = 2, \dots, 7$  at  $L \leq 10$  and  $d = 2, 3$  at  $L = 12$ ;  $p = 0$  ranks are exact over  $\mathbb{Z}$  for  $L \leq 8$  ( $d = 2, 3$ ) and modular over two primes at  $L = 10, 12$ ;  $p_{\times}$  ranks are modular over two primes for  $d = 3, \dots, 7$  ( $L \leq 10$ ) and for  $d = 3$  at  $L = 12$  (primes 1048261 and 1048573, different seeds, identical block ranks). At  $p = 1$  the rank is 1 for every  $L$  and  $d$  (Theorem 5(iii)).

| $L$ | $6^{L/2}$ | $r_3(L)$ | generic | $p = 0$ | $p_{\times}$ | $ E_L , d = 2/3$ |
|-----|-----------|----------|---------|---------|--------------|------------------|
| 4   | 36        | 21       | 21      | 21      | 21           | 0 / 1            |
| 6   | 216       | 216      | 216     | 198     | 176          | 1 / 6            |
| 8   | 1296      | 1202     | 1202    | 1184    | 1142         | 11 / 55          |
| 10  | 7776      | 7776     | 7776    | 7663    | 7276         | ? / ?            |
| 12  | 46656     | 46080    | 46010   | 45822   | 44898        | ? / ?            |

-c `certificate_sha256.txt` verifies the deposit. The later deposit rounds extend the ledger without altering any entry above: `certificate_sha256_v5.txt` (the  $n = 4$  colour-resolved scans, the first  $n = 5$  diagnostics, and the Haar sequential-Monte-Carlo results quoted in the main text), `certificate_sha256_v6.txt` (the  $n = 5$  two-size test, the  $n = 4$   $L = 10$  third crossing, and the no-freezing-theorem verification), `certificate_sha256_v8.txt` (the v21 round: the manuscript and supplement of that round together with the Clifford record-count suite whose runs are tabulated in Sec. SVII; the parallel-line ledger of that suite is `certificate_sha256_v3_scgf.txt`), and `certificate_sha256_v9.txt` (the v22 round: the exact-closure suite of Sec. SVIID together with the `gap_utils` copies that repair the reproduction chain of the v17–v19 scripts), each self-verifying in the same way. Table S9 lists the scripts (Python 3, dependencies `numpy` and `python-flint`; `sympy` for the  $d = 2$  number-field ranks). The reproduction sequence is given in Table S7.

The  $L = 12$  lower-bound runs take about 15 minutes each on one core with less than 1.5 GB of memory; all other steps take seconds to a few minutes.

## SV. PROOF OF THE FREEZING THEOREM (THEOREM 8 OF THE MAIN TEXT)

The main text states the theorem in Sec. IV and defers the proof here. The statement is reproduced with equations renumbered. Let  $a = \log 2$  and  $s_c = \sqrt{2a} = 1.17741$ . On one probability space let  $\{E_{T,\omega}(Y) : T \geq 1, Y \in \{0, 1\}^T\}$  be independent Gaussian variables with  $E_{T,\omega}(Y) \sim \mathcal{N}(0, T)$ , and for real  $s$  put

$$\mathcal{Z}_{s,T,\omega} = \sum_{Y \in \{0,1\}^T} e^{-sE_{T,\omega}(Y)}, \quad P_{T,\omega}(Y) = \frac{e^{-E_{T,\omega}(Y)}}{\mathcal{Z}_{1,T,\omega}}. \quad (\text{S1})$$

(i) (Born realizability.) For every  $T$  and  $\omega$ ,  $P_{T,\omega}$  is the record law of the sequential computational-basis measurement of the  $T$  qubits of the pure state  $|\Psi_{T,\omega}\rangle = \sum_Y \sqrt{P_{T,\omega}(Y)} |Y\rangle$ , and the predictors of Definition 1 of the main text are  $r_{T,\omega,t}(y_t | y_{<t}) = P_{T,\omega}(y_{\leq t})/P_{T,\omega}(y_{<t})$ , so that  $\mathcal{A}_T = -\log P_{T,\omega}(Y)$  and  $\phi_{T,\omega}(k) = T^{-1} \log \sum_Y P_{T,\omega}(Y)^{1-k}$  are exactly the sample log-MGFs of that definition.

(ii) (Extremes.) As  $T \rightarrow \infty$ , almost surely,  $\max_Y (-E_{T,\omega}(Y))/T \rightarrow s_c$  and  $\min_Y (-E_{T,\omega}(Y))/T \rightarrow -s_c$ .

(iii) (Quenched free energy.) Almost surely, simultaneously for all real  $s$ ,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log \mathcal{Z}_{s,T,\omega} = \varphi(s) = \begin{cases} a + s^2/2, & |s| \leq s_c, \\ |s| s_c, & |s| \geq s_c, \end{cases} \quad (\text{S2})$$

and the convergence is locally uniform in  $s$ ;  $\varphi$  is convex and  $C^1$  on  $\mathbb{R}$ ,  $\varphi''$  jumps from 1 to 0 at  $\pm s_c$ , and  $\varphi(1) = a + \frac{1}{2}$  because  $1 < s_c$ .

(iv) (Record exponents.) Almost surely, for every real

TABLE S7. Reproduction commands (Python 3; NB = 2, 3, 4 is the number of bonds,  $L = 2\text{NB}$ ).

| command                                                                   | purpose                                                       |
|---------------------------------------------------------------------------|---------------------------------------------------------------|
| <code>python3 n3_mult_certificates.py NB</code>                           | $d = 2$ interval certificates (Table II of the main text)     |
| <code>python3 n3_mult_certificates_d.py NB 3</code>                       | $d = 3$ interval certificates                                 |
| <code>python3 n3_exceptional_exact.py NB</code>                           | $d = 2$ exact ranks at the exceptional points (Table S1)      |
| <code>python3 n3_exceptional_exact_d.py NB 3</code>                       | $d = 3$ exact ranks at the exceptional points (Tables S2, S3) |
| <code>python3 n3_simple_zero_check.py</code>                              | order of vanishing of the certificate factors                 |
| <code>python3 n3_p0_exact_rank.py</code>                                  | $p = 0$ ranks over $\mathbb{Z}$ , $L \leq 8$                  |
| <code>python3 n3_closed_form_proof.py</code>                              | closed form of the Burnside deficit, Eq. (rank3)              |
| <code>python3 n3_sharpness_L12.py 6 [--d D --p P --q Q --seed S]</code>   | modular lower bounds, all blocks (Table S5)                   |
| <code>python3 n3_L12_structure_check.py</code>                            | antisymmetry of the $k = m/2$ sector                          |
| <code>python3 n3_L12_kernel.py 6</code>                                   | constant kernel vectors ( $k = 3$ )                           |
| <code>python3 n3_L12_kernel_verify.py 6</code>                            | exact verification and all- $p$ upper bound                   |
| <code>python3 n3_L12_moving_kernel.py 6 --block B --side - [--d 3]</code> | polynomial kernel vectors                                     |
| <code>python3 n3_L12_final.py [--d 3]</code>                              | two-sided certificate (Table S4)                              |
| <code>python3 n3_kernel_recheck.py [--d 3]</code>                         | independent re-verification of all kernel vectors             |

$k$  and every  $q \geq 0$ ,

$$\psi(k) = \lim_{T \rightarrow \infty} \phi_{T,\omega}(k) = \varphi(1-k) - (1-k)\varphi(1), \quad (\text{S3})$$

$$\begin{aligned} \tau(q) &= -\psi(1-q) = q\varphi(1) - \varphi(q) \\ &= \begin{cases} q(a + \frac{1}{2}) - a - \frac{q^2}{2}, & 0 \leq q \leq q_c, \\ q(a + \frac{1}{2} - s_c), & q \geq q_c, \end{cases} \end{aligned} \quad (\text{S4})$$

with  $q_c = s_c$ . Thus  $\tau$  is  $C^1$  on  $[0, \infty)$  with a linear branch of slope  $a + \frac{1}{2} - s_c = 0.01574$  for  $q \geq q_c$ ,  $\tau''$  jumps from  $-1$  to  $0$  at  $q_c$ , and  $\psi$  is convex, finite and  $C^1$  on  $\mathbb{R}$  but not analytic at  $k_c = 1 - s_c = -0.17741$  (and, on the unphysical side  $q < 0$ , at  $1 + s_c = 2.17741$ ). Explicitly  $\psi(k) = \frac{1}{2}k^2 + (a - \frac{1}{2})k$  for  $k_c \leq k \leq 1 + s_c$  and  $\psi$  is affine outside this interval.

- (v) (Large deviations.)  $\psi$  is essentially smooth, so the main-text Theorem 7 applies to the sequence  $(P_{T,\omega})_T$ : the surprisal rate  $h = \mathcal{A}_T/T$  satisfies the full LDP with the good rate function  $I(h) = \frac{1}{2}(h - h^*)^2$  for  $h \in [h_{\min}, h_{\max}]$  and  $I(h) = +\infty$  otherwise, where  $h^* = \psi'(0) = a - \frac{1}{2} = 0.19315$  is the typical surprisal rate,  $h_{\min} = \psi'(k_c) = a + \frac{1}{2} - s_c = 0.01574$  and  $h_{\max} = \psi'(1 + s_c) = a + \frac{1}{2} + s_c = 2.37056$ . The singularity spectrum is  $f(\alpha) = \alpha - I(\alpha)$  on  $[h_{\min}, h_{\max}]$ , with  $f(h_{\min}) = 0$ : the most probable records,  $P_{T,\omega}(Y) = e^{-Th_{\min} + o(T)}$ , are subexponentially many, and the tilted ensembles  $P_{q,\omega}$  with  $q > q_c$  all concentrate on them.

*Proof.* (i) The state is normalized because  $\sum_Y P_{T,\omega}(Y) = 1$ . Measuring qubit 1 in the computational basis gives  $y_1$  with probability  $\sum_{y_{>1}} P_{T,\omega}(y_1 y_{>1}) = P_{T,\omega}(y_1)$  and leaves the normalized state  $\sum_{y_{>1}} \sqrt{P_{T,\omega}(y_1 y_{>1})/P_{T,\omega}(y_1)} |y_1 y_{>1}\rangle$ ; inductively the Born predictor at step  $t$  is  $P_{T,\omega}(y_{\leq t})/P_{T,\omega}(y_{< t})$  and the product of the predictors telescopes to  $P_{T,\omega}(Y)$ . The

process is a monitored quantum process in the sense of Sec. IV of the main text with the post-measurement state as  $X_{t-1}$  and no unitary steps; it is nonlocal in the record index and its Hilbert space has dimension  $2^T$ .

- (ii) Write  $g_Y = -E_{T,\omega}(Y)/\sqrt{T}$ , i.i.d. standard Gaussians,  $M_T = \max_Y g_Y$ , and  $\bar{\Phi}(x) = \Pr(g > x)$ . Upper bound: for  $\varepsilon > 0$ , by the union bound and  $\bar{\Phi}(x) \leq e^{-x^2/2}$ ,

$$\Pr(M_T > \sqrt{T}(s_c + \varepsilon)) \leq 2^T e^{-T(s_c + \varepsilon)^2/2} = e^{-T(\varepsilon s_c + \varepsilon^2/2)},$$

using  $2^T = e^{Ts_c^2/2}$ ; the right side is summable in  $T$ . Lower bound: by independence and  $1 - x \leq e^{-x}$ ,

$$\begin{aligned} \Pr(M_T < \sqrt{T}(s_c - \varepsilon)) &= (1 - \bar{\Phi}(\sqrt{T}(s_c - \varepsilon)))^{2^T} \\ &\leq \exp(-2^T \bar{\Phi}(\sqrt{T}(s_c - \varepsilon))), \end{aligned}$$

and the Gaussian tail bound  $\bar{\Phi}(x) \geq \frac{x}{1+x^2} \frac{e^{-x^2/2}}{\sqrt{2\pi}}$  gives  $2^T \bar{\Phi}(\sqrt{T}(s_c - \varepsilon)) \geq cT^{-1/2} e^{T(\varepsilon s_c - \varepsilon^2/2)}$  for a constant  $c > 0$ , which grows exponentially for  $0 < \varepsilon < 2s_c$ , so the probability is summable. By the Borel-Cantelli lemma  $M_T/\sqrt{T} \rightarrow s_c$  almost surely, i.e.  $T^{-1} \max_Y (-E_{T,\omega}(Y)) \rightarrow s_c$ ; the minimum follows from the symmetry  $E \mapsto -E$ .

- (iii) Fix  $s > 0$  (the case  $s < 0$  follows from  $E \mapsto -E$ , and  $s = 0$  is  $\mathcal{Z}_0 = 2^T$ ). *Frozen upper bound.*  $\mathcal{Z}_{s,T,\omega} \leq 2^T e^{s \max_Y (-E)}$ , so by (ii), almost surely  $\limsup_T T^{-1} \log \mathcal{Z}_{s,T,\omega} \leq a + ss_c$ . *Annealed upper bound.*  $\mathbb{E} \mathcal{Z}_{s,T,\omega} = 2^T e^{s^2 T/2}$ , so by Jensen's inequality  $\mathbb{E} \log \mathcal{Z}_{s,T,\omega} \leq T(a + s^2/2)$ . To pass from the mean to the sample, note that  $F(E) = \log \sum_Y e^{-sE(Y)}$  has gradient  $-\pi_s$  with  $\pi_s$  a probability vector, so  $F$  is  $|\pi_s|$ -Lipschitz for the Euclidean norm of  $E \in \mathbb{R}^{2^T}$ , and  $E = \sqrt{T}G$  with  $G$  a standard Gaussian vector; the Gaussian concentration inequality of Tsirelson, Ibragimov and Sudakov (Ref. [1], Theorem 5.6, applied to the  $|\pi_s|\sqrt{T}$ -Lipschitz

function  $G \mapsto F(\sqrt{T}G)$  gives

$$\Pr(|\log \mathcal{Z}_{s,T,\omega} - \mathbb{E} \log \mathcal{Z}_{s,T,\omega}| > \varepsilon T) \leq 2e^{-\varepsilon^2 T / (2s^2)}, \quad (\text{S5})$$

which is summable; hence almost surely  $\limsup_T T^{-1} \log \mathcal{Z}_{s,T,\omega} \leq a + s^2/2$ . Together,  $\limsup \leq \min(a + s^2/2, a + ss_c)$ ; this is  $\varphi(s)$  for  $s \leq s_c$  but exceeds  $\varphi(s) = ss_c$  for  $s > s_c$ , and the sharp value comes from the lower bound. *Lower bound by truncation.* Fix  $0 < e < s_c$  and  $\delta > 0$  with  $e + \delta < s_c$ , and let  $N_T$  be the number of records with  $-E_{T,\omega}(Y) \in [Te, T(e + \delta)]$ , a binomial variable with  $2^T$  trials and success probability  $\pi_T = \bar{\Phi}(\sqrt{T}e) - \bar{\Phi}(\sqrt{T}(e + \delta)) \geq c'T^{-1/2}e^{-T(e+\delta)^2/2}$  for large  $T$  and some  $c' > 0$ . Then  $\mathbb{E}N_T = 2^T \pi_T \geq c'T^{-1/2}e^{T(a-(e+\delta)^2/2)}$  grows exponentially, and Chebyshev's inequality gives  $\Pr(N_T < \frac{1}{2}\mathbb{E}N_T) \leq 4\text{Var } N_T / (\mathbb{E}N_T)^2 \leq 4/\mathbb{E}N_T$ , which is summable. Almost surely, therefore, for all large  $T$ ,  $\mathcal{Z}_{s,T,\omega} \geq N_T e^{sTe} \geq \frac{1}{2}\mathbb{E}N_T e^{sTe}$ , i.e.  $\liminf_T T^{-1} \log \mathcal{Z}_{s,T,\omega} \geq a - (e + \delta)^2/2 + se$ . Letting  $\delta \downarrow 0$  along a sequence and taking the supremum over rational  $e \in (0, s_c)$  (countably many events, each of full probability),

$$\begin{aligned} \liminf_T \frac{1}{T} \log \mathcal{Z}_{s,T,\omega} &\geq \sup_{0 \leq e \leq s_c} \left( a - \frac{e^2}{2} + se \right) \\ &= \begin{cases} a + s^2/2, & s \leq s_c, \\ a - s_c^2/2 + ss_c = ss_c, & s \geq s_c, \end{cases} \end{aligned}$$

because the concave function  $e \mapsto a - e^2/2 + se$  is maximized at  $e = s$  when  $s \leq s_c$  and at the endpoint  $e = s_c$  otherwise, and  $a = s_c^2/2$ . Upper and lower bounds agree, which proves Eq. (S2) for each fixed  $s$  on an event of probability one. *Uniformity.* Each  $s \mapsto T^{-1} \log \mathcal{Z}_{s,T,\omega}$  is convex (a log-sum-exp), so convergence on the countable dense set  $\mathbb{Q}$  to the finite convex function  $\varphi$  implies convergence for every real  $s$ , locally uniformly (Ref. [2], Theorem 10.8); the intersection of countably many full-probability events has probability one. The stated regularity of  $\varphi$  is read off from Eq. (S2): the two branches and their first derivatives ( $s$  and  $s_c \text{sgn } s$ ) agree at  $|s| = s_c$ , the second derivatives do not.

(iv) By Eq. (S1),  $\sum_Y P_{T,\omega}(Y)^{1-k} = \mathcal{Z}_{1-k,T,\omega} / \mathcal{Z}_{1,T,\omega}^{1-k}$ , so  $\phi_{T,\omega}(k) = T^{-1} \log \mathcal{Z}_{1-k,T,\omega} - (1-k)T^{-1} \log \mathcal{Z}_{1,T,\omega}$ , and (iii) gives Eq. (S3) on the same full-probability event, simultaneously for all  $k$ ; Eq. (S4) is the identity  $\tau(q) = -\psi(1-q)$  of Theorem 7 of the main text with  $\varphi(1) = a + \frac{1}{2}$  and Eq. (S2) at  $s = q \geq 0$ . The regularity statements follow from those of  $\varphi$ ; in particular  $\tau'(q) = a + \frac{1}{2} - q$  on  $[0, q_c]$  and  $\tau'(q) = a + \frac{1}{2} - s_c$  on  $[q_c, \infty)$ , continuous at  $q_c$ , and  $\tau'' = -1$ , respectively 0. The quadratic form of  $\psi$  on  $[k_c, 1 + s_c]$  is  $\varphi(1-k) - (1-k)\varphi(1) = a + \frac{1}{2}(1-k)^2 - (1-k)(a + \frac{1}{2}) = \frac{1}{2}k^2 + (a - \frac{1}{2})k$ .

(v)  $\psi$  is finite, convex and differentiable on all of  $\mathbb{R}$ , hence essentially smooth (Ref. [3], Definition 2.3.5), and 0 is interior to its domain, so all hypotheses of Theorem 7 of the main text hold for the sequence  $(P_{T,\omega})_T$ , whose

finite-horizon log-MGFs are  $\phi_{T,\omega}$  by (i). The Legendre transform of the quadratic branch  $\frac{1}{2}k^2 + (a - \frac{1}{2})k$  is  $\frac{1}{2}(h - h^*)^2$  with  $h^* = a - \frac{1}{2}$ , attained at  $k = h - h^*$ , which lies in  $[k_c, 1 + s_c]$  exactly when  $h \in [h_{\min}, h_{\max}]$  with  $h_{\min} = k_c + h^* = a + \frac{1}{2} - s_c$  and  $h_{\max} = 1 + s_c + h^* = a + \frac{1}{2} + s_c$ . On the affine branches  $\psi(k) = h_{\min}k + (\psi(k_c) - h_{\min}k_c)$  for  $k < k_c$  and  $\psi(k) = h_{\max}k + (\psi(1 + s_c) - h_{\max}(1 + s_c))$  for  $k > 1 + s_c$  (the slopes are  $\psi'(k_c)$  and  $\psi'(1 + s_c)$  by continuity of  $\psi'$ ), so  $\sup_k [kh - \psi(k)] = +\infty$  for  $h < h_{\min}$  and for  $h > h_{\max}$ . This gives  $I$  as stated and, by  $f(\alpha) = \alpha - I(\alpha)$  (Theorem 7 of the main text),  $f(\alpha) = \alpha - I(\alpha)$ ;  $f(h_{\min}) = h_{\min} - \frac{1}{2}(h_{\min} - h^*)^2 = h_{\min} - \frac{1}{2}k_c^2$ , and indeed  $h_{\min} - \frac{1}{2}k_c^2 = (a + \frac{1}{2} - s_c) - \frac{1}{2}(1 - s_c)^2 = a - \frac{1}{2}s_c^2 = 0$ . The concentration statement is the meaning of  $f(h_{\min}) = 0$  together with Eq. (S4): for  $q > q_c$ ,  $\tau(q) = qh_{\min}$ , so  $Z_{q,\omega}(T) = e^{-Tqh_{\min} + o(T)}$  is carried by the records of minimal surprisal rate.  $\square$

## SVI. PROOF OF THE FINITE-POPULATION NORMALIZATION PROPERTY (PROPOSITION OF THE MAIN TEXT)

The main text defines the bootstrap estimator for the record SCGF and states its finite-population normalization property, deferring the proof here. For a fixed time-homogeneous environment with outcome kernel  $K_\omega(x, dy)$ , update  $F_\omega(x, y)$  and additive increment  $g_\omega(x, y)$  (in the record application,  $g = -\log r$ ), the Feynman-Kac operator of the main text is

$$(\mathcal{Q}_{k,\omega}f)(x) = \int K_\omega(x, dy) e^{kg_\omega(x,y)} f(F_\omega(x, y)). \quad (\text{S6})$$

The estimator: initialize  $N$  normalized conditional states; at event  $t$ , independently for each particle  $i$ , sample  $Y_t^i$  from its Born predictor, update and normalize the state, and set  $\ell_t^i = -k \log r_t^i$  (for a compound block,  $r_t^i$  is the product of its sequential conditional Born probabilities and  $-\log r_t^i$  their sum). With resampling at every declared event, compute

$$m_t = \max_i \ell_t^i, \quad \log c_t = m_t + \log \left( \frac{1}{N} \sum_i e^{\ell_t^i - m_t} \right), \quad \bar{w}_t^i = \frac{e^{\ell_t^i - m_t}}{\sum_j e^{\ell_t^j - m_t}}$$

then resample with a conditionally unbiased scheme and reset weights. The estimates are

$$\hat{Z}_T(k) = \prod_{t=1}^T c_t, \quad \hat{\psi}_{L,\omega}(k) = \frac{1}{T} \sum_{t=1}^T \log c_t. \quad (\text{S7})$$

Under conditionally unbiased resampling,  $\hat{Z}_T(k)$  in Eq. (S7) is unbiased for  $Z_T(k) = \mathbb{E}_{P_\omega} e^{kA_T}$  at every finite  $N, T$  [4]; in general  $\log \hat{Z}_T$  is biased; and if outcomes are drawn from an auxiliary proposal  $Q_t$  rather than the Born kernel  $K_t$ , the incremental weight must be

$$w_t = e^{kg(x,y)} \frac{K_t(x, dy)}{Q_t(x, dy)}. \quad (\text{S8})$$

*Proof.* Let  $\gamma_t(f) = \mathbb{E}_{P_\omega}[f(X_t)e^{kA_t}]$  be the unnormalized Feynman–Kac measure, so that  $Z_t(k) = \gamma_t(1)$  and  $\gamma_t(f) = \gamma_{t-1}(\mathcal{Q}_{k,\omega}f)$  with the operator of Eq. (S6) (the increment  $g = -\log r$  gives  $e^{kg} = e^\ell$ ). Let  $\eta_t^N = N^{-1} \sum_i \delta_{x_t^i}$  be the equally weighted particle measure after resampling at event  $t$ ,  $\hat{\eta}_t^N = \sum_i \bar{w}_t^i \delta_{x_t^i}$  the weighted one before resampling, and  $\mathcal{F}_{t-1}$  the history up to the resampling at event  $t-1$ . Because particle  $i$  draws  $Y_t^i$  from its own Born predictor independently of the others,  $\mathbb{E}[e^{\ell_i} f(x_t^i) \mid \mathcal{F}_{t-1}] = (\mathcal{Q}_{k,\omega}f)(x_{t-1}^i)$ ; averaging over  $i$  and using  $c_t \hat{\eta}_t^N(f) = N^{-1} \sum_i e^{\ell_i} f(x_t^i)$  gives  $\mathbb{E}[c_t \hat{\eta}_t^N(f) \mid \mathcal{F}_{t-1}] = \eta_{t-1}^N(\mathcal{Q}_{k,\omega}f)$ . Conditionally unbiased resampling means  $\mathbb{E}[\eta_t^N(f) \mid \hat{\eta}_t^N] = \hat{\eta}_t^N(f)$ . Multiplying by the  $\mathcal{F}_{t-1}$ -measurable  $\hat{Z}_{t-1} = \prod_{s < t} c_s$  and combining the two conditional expectations,

$$\mathbb{E}[\hat{Z}_t \eta_t^N(f)] = \mathbb{E}[\hat{Z}_{t-1} \eta_{t-1}^N(\mathcal{Q}_{k,\omega}f)].$$

By induction from  $\hat{Z}_0 = 1$  and  $\eta_0^N = \delta_{x_0} = \gamma_0$  this gives  $\mathbb{E}[\hat{Z}_t \eta_t^N(f)] = \gamma_0(\mathcal{Q}_{k,\omega}^t f) = \gamma_t(f)$  for every bounded  $f$ , and  $f = 1$  yields  $\mathbb{E}\hat{Z}_T = Z_T$  (Ref. [4], Theorem 7.4.2). Jensen’s inequality gives  $\mathbb{E} \log \hat{Z}_T \leq \log Z_T$ , with equality only if  $\hat{Z}_T$  is almost surely constant, so the logarithm is biased in general. Equation (S8) is the Radon–Nikodym change-of-measure identity: with outcomes drawn from  $Q_t$ , the conditional expectation of  $w_t f(x')$  is  $\int Q_t(x, dy) e^{kg(x,y)} \frac{K_t(x, dy)}{Q_t(x, dy)} f(F(x, y)) = (\mathcal{Q}_{k,\omega}f)(x)$ , and without the ratio the proposal expectation is not the target Feynman–Kac integral.  $\square$

## SVII. THE CLIFFORD RECORD-COUNT RUNS

This section documents the runs behind the main-text Sec. IV.F (the disorder SCGF of the Clifford record count): the exact evolution and its calibration, the trajectory simulator and its validation, the production ladder, the estimator definitions, and the reproduction commands.

### A. The exact $3^L$ algebra evolution and the deposit calibration

The annealed quantities are computed exactly by evolving the two-replica state on the per-site algebra  $\{e_0, e_+, e_-\}^{\otimes L}$  (the  $3^L$  product basis): the gate-bond twirl is the exact factorised projection  $\alpha I + \beta S$  with  $\alpha = (4t_i t_j - s_i s_j)/60$ ,  $\beta = (4s_i s_j - t_i t_j)/60$  ( $d = 2$ ; per-site  $(t, s)$ :  $e_0$ : (2, 2),  $e_+$ : (1, 1),  $e_-$ : (1, -1)), the measurement layer is diagonal ( $e_0 \mapsto e_0$ ,  $e_\pm \mapsto (1-p)e_\pm$ : the measure/no-measure choice is a single event shared by both replicas—the naive  $(1-p)^2$  damping was caught and corrected by the deposit calibration below), and the initial state  $|0 \cdots 0\rangle\langle 0 \cdots 0|^{\otimes 2}$  maps, after the (inert) first measurement layer and the first gate layer, to

$\otimes_{\text{even bonds}} [(e_0 + e_+)^{\otimes 2} + e_-^{\otimes 2}]/10$  (trace 1 exactly). The observables are linear functionals:  $Z_2 = \sum_a c_a \prod_k t_{a_k}$  and  $P_{2,A} = \sum_a c_a \prod_k t_{a_k} \prod_{k \in A} \varepsilon_{a_k}$  with  $\varepsilon = (1, 1, -1)$ , the last being the exact algebraic form of the replica swap  $S_A$ .

The calibration (Table S10) fixes the deposited chain’s time convention: the deposited chain runs at  $t = L/2$  periods ( $t = 4$  at  $L = 8$ ,  $t = 6$  at  $L = 12$ ); the deposited log’s own “ $t = 4L$ ” reading of this chain is a mis-transcription (at  $t = 4L = 32$  one instead gets  $Z_2 = 4.57 \times 10^{-18}$ , twenty-nine orders away). The deposited  $L = 12$  quenched entropy 2.394(5) is not reproduced at either convention (our chains: 2.497(7) at  $t = 4$ , 2.632(5) at  $t = 6$ ) while both deposited annealed values are matched at  $t = 6$ ; it is quoted in the main-text Appendix B as deposited and flagged here as a provenance exception on which nothing in the main text depends. The trajectory ensemble is an independent implementation (vectorised phase-free stabilizer tableau, uint64  $x/z$  words, the 720 two-qubit Clifford actions as lookup tables, M–A–M–B period order, new PCG64 seed contract `seed0 = 20260529 + cell`) and reproduces both the exact  $Z_2$  (relative deviation 0.73% and 0.59% at  $L = 8, 12$ ) and the deposited quenched chain at  $L = 8$ .

### B. Exact anchors and the production ladder

The  $\beta = 1$  trajectory estimates degrade with the ESS collapse (e.g.  $-0.08037$  at  $L = 8$  against the exact  $-0.07798$ , the bias growing with  $L$ ), which is why Table S12 reports the exact anchors in the  $\Lambda_L(1)$  column; the trajectory columns are used only for  $\beta \leq 0.75$ , where the moment estimates are well conditioned. The  $\Lambda_L(2)$  trajectory estimates ( $L = 8, 12, 16$  at  $p = 0.16$ :  $-0.14486$ ,  $-0.15891$ ,  $-0.17019$ ; at  $p = 0.22$ :  $-0.20652$ ,  $-0.22329$ ,  $-0.23247$ ) underlie the ESS law of the main text;  $\Lambda_L(2)$  is exactly the  $n = 3$  annealed moment  $(2Lt)^{-1} \ln \bar{Z}_3(t)$  of the main-text Eq. (dyadic moments) and is in principle computable exactly by the three-replica operator.

### C. Estimators, error model, and reproduction

The trajectory estimator of  $\Lambda_L(\beta)$  at fixed  $(L, p, t)$  is the logarithm of the sample mean of  $2^{-\beta N_b}$  over  $B$  independent disorder draws  $b$  (each draw contributing one record under its own Born measure;  $N_b = N_{\text{rand}}(\omega_b)$  is  $\omega_b$ -measurable, so one record per draw suffices), with bootstrap standard errors over 200 trajectory resamples; the tilted densities are the weighted means  $\tilde{m}(\beta) = \sum_b 2^{-\beta N_b} N_b / \sum_b 2^{-\beta N_b} / (2Lt)$ ; the ESS is the within-sample ratio  $(\sum_b w_b)^2 / \sum_b w_b^2$ . The raw arrays ( $X_R$  and  $S_{L/2}$  per trajectory, all eight production cells) are deposited as `scgf_born_raw_L{8,12,16,24}_p{16,22}.npz`.

Reproduction (from the `research/scripts/` directory):

| command                                 | purpose                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------|
| <code>python3 mipt_scgf_exact.py</code> | the $3^L$ evolution: calibration, anchors, $4(L)$ Ail/Tables (S10, S11)                |
| <code>python3 mipt_born_scgf.py</code>  | the trajectory suite: validation, production ladder, ESS, tilted densities (Table S12) |

Dependencies: `numpy` (PCG64). Wall time: the full suite is minutes on one core except the  $L = 16$  exact evolution ( $3^{16}$  configurations, hours). The files of this round are hashed in `certificate_sha256_v8.txt` (round files) and `certificate_sha256_v3_scgf.txt` (the parallel-line ledger of the record-SCGF suite itself).

#### D. The v22 exact-closure runs

*a. The reconstructed library.* The shared module `gap_utils.py` (the Weingarten channel  $W_{p,n}$  of Eq. (Wpn), the group utilities, and the dense bond-label operators `bond_ops`) on which the deposited v17–v19 script suites all depend was missing from the repository: the deposit’s reproduction chain was silently broken. It is rebuilt exactly from Eq. (Wpn) (inverse of the Gram  $D^{c(\sigma^{-1}\tau)}$  for  $D \geq n$ , Moore–Penrose pseudo-inverse otherwise), placed both in `scripts/v22-exactZ3-n5L8/` and as copies into `v17-annealed-n3/`, `v18-n4n5-smc/` and `v19-n5L6-nofreeze/`, and validated by re-running the deposited batteries unmodified: `n3_annealed_validate_v1.py` V1–V6 all PASS (the Sec. 9 benchmarks to  $0.0 \times 10^0$  at six decimals, the Gram similarity  $1.7 \times 10^{-16}$ , the Kaufman closed form to  $4 \times 10^{-11}$ , the momentum-block union  $1.3 \times 10^{-16}$ , the colour sectors to  $2 \times 10^{-6}$ , the product rows exact) and `n45_annealed_validate_v1.py` V1–V8 ALL OK (including the  $n = 5$   $p = 1$  bond eigenvalue  $1/14$  through the pseudo-inverse channel and the pinv property  $G \text{Wg} G = G$  to  $3 \times 10^{-15}$ ).

*b. The conjugation-design test.* `v22_clifford_3design_v1.py` enumerates the two-qubit Clifford group by BFS closure from  $\{H \otimes I, I \otimes H, S \otimes I, I \otimes S, \text{CNOT}_{01}, \text{CNOT}_{10}\}$ : 92,160 unitaries, the phase extension  $\{e^{ik\pi/4}\} \times \text{Cl}_2$ ; phase classes are deduplicated by the injective phase-invariant key  $U \mapsto U \otimes \bar{U}$  (11,520 classes), and the census of the unsigned-Pauli actions gives exactly the 720 symplectic classes with 16 elements each. The test averages  $C^{\otimes n} X C^{\dagger \otimes n}$  over the classes and compares with the Haar twirl  $\sum_{\pi} U_{\pi} y_{\pi}$ ,  $\text{Gy} = t$ ,  $\text{G}[\pi, \rho] = (d^2)^{c(\pi^{-1}\rho)}$ , on six random  $X$  at each  $n$ :  $n = 2$  deviates  $2.1 \times 10^{-14}$ ,  $n = 3$  deviates  $1.6 \times 10^{-14}$  (the identification of Remark `rem:cliff3design`),  $n = 4$  deviates 0.578 (the negative control).

*c. The exact  $\bar{Z}_3$  cells and ladder.* `v22_z3_exact_v1.py` implements the boundary-vector formula of Prop. `prop:z3closure` on the compressed bond space with the `ring_gemm_v2` kernels. Validation: the  $n = 2$  anchors of `scgf_exact.json` to  $2.6 \times 10^{-15}$ , the  $A$ -tail  $2.4509 \rightarrow 2.4512$ , the layer-parity invariance to 0.0. Production cells ( $t = 4L$ ): Table `tab:z3exact` of the

main text. The ladder:  $\lambda_1^{(3)}$  by power iteration (validated against the dense  $L = 8$  spectrum to  $1.9 \times 10^{-10}$ ), with amplitudes  $\text{Ail}(L)$  Tables (S10, S11) and subexponential in  $t$ .  
*d. The  $\beta = 2$  identification cross-check.*

`v22_traj_beta2_v1.py` runs the deposited tableau simulator at  $t = L/2$  with  $B = 4 \times 10^6 / 2.4 \times 10^6 / 1.6 \times 10^6$  trajectories at  $L = 8/12/16$  (fresh PCG64 seeds, recorded):  $\hat{\mathbb{E}}[2^{-2N_{\text{rand}}}]$  against the exact  $\bar{Z}_3(t)$  and  $\hat{\mathbb{E}}[2^{-N_{\text{rand}}}]$  against  $\bar{Z}_2(t)$ , with bootstrap standard errors and the empirical tilted ESS. The results are quoted in the main text (Prop. `prop:z3closure`).

*e. The marginal  $q = 4$  log-correction fits.* `v22_n4_eps_topup_v1.py` recomputes the missing  $L = 8$   $\lambda_{\epsilon}$  values ( $p = 0.39, 0.40, 0.41$ ; the deposited  $p = 0.38$  value reproduced to  $5.7 \times 10^{-16}$  first). `v22_q4_logcorr_v1.py` then evaluates  $R_L$ , the slopes and the gaps at the common reference  $p^* = 0.383$ , and fits the marginal forms:  $R_L = 1/4 - 0.90/\ln L + 0.46/\ln^2 L$  (residual  $\leq 8 \times 10^{-4}$ , against the corrected  $q = 4$  target  $x_{\sigma}/x_{\epsilon} = 1/4$ );  $dX_L/dp = C L^{3/2}(1 + 0.45/\ln L)$ ; the  $n = 3$  control converging cleanly to  $1/6$ . The crossing table is recomputed with the top-up ( $p_c$  crossings  $0.3582/0.3790/0.3823$ ;  $1/\nu_{\text{eff}} 1.00 \rightarrow 1.38 \rightarrow 1.57$ ).

*f. The tilted-variance profile family.* `v22_tiltvar_uniform_v1.py` measures  $\text{Var}_s(\log X)$  on  $s \in [0, 1.5]$  (61 points, tilted empirically, 200 bootstrap resamples): Part A, exactly enumerated Haar  $p = 1$  collision partition functions ( $k^* = -1$ ) over the grid  $L = 3-6$ ,  $T = 2L, 4L, 8L$ , 6000 circuits per cell; Part B, fresh Clifford disorder trajectories at  $t = L/2$  (the same well-conditioned cells as the  $\beta = 2$  cross-check). Metrics per cell: the per-period variance rate  $v(0) = \text{Var}_0/T$  (T-independent within each  $L$  to 3%), the sub-Gaussian profile ratio  $\rho = 2 \int_0^1 (1-s)v(s)ds/v(0)$ , the initial decay rate  $\kappa$ , and the interpolation identity as a per-cell validation ( $\leq 2.5 \times 10^{-6}$ ). Part A:  $\rho = 1.02 \pm 0.03$  across the whole grid (a flat-to-mildly-rising profile,  $(L, T)$ -uniform in shape); Part B (Clifford disorder):  $\rho = 0.872/0.878/0.888$  at  $L = 8/12/16$ ,  $p = 0.16$  ( $0.890/0.894/0.897$  at  $p = 0.22$ ): the sub-Gaussian decaying profile of the v21 Discussion, confirmed on well-conditioned data and mildly de-sub-Gaussianizing with  $L$ . The raw production  $t = 4L$  arrays are NOT used for Part B: their tilted ESS collapses to 1, the same estimator difficulty as the  $\beta = 2$  moment (the within-sample profiles are reported in the JSON for completeness, flagged).

*g. The symmetry-reduced  $n = 5$   $L = 8$  block route.* `v22_n5_L8_block_v1.py` assembles the momentum-zero colour-trivial block of  $\mathbb{C}_{\text{comp}}^{(5)}$  at  $L = 8$  exactly: the orbit basis of the symmetry group  $(S_5 \times S_5) \rtimes \mathbb{Z}_4$  on  $(S_5)^4$  (canonical forms = the minimum over conjugators and shifts of the encoded lkey tuple; the orbit table is  $p$ -independent and memmapped), the block elements  $B[\alpha, \beta] = \sqrt{n_{\alpha}/n_{\beta}} \sum_{\tau \in O_{\beta}} M[\text{rep}_{\alpha}, \tau]$  by a chunked pass (segment sums over orbit-sorted chunks), and

$C_{\text{block}} = B_{M_1} B_{M_2}$ . Validation: the deposited  $n = 5$   $L = 4$  `triv3` rows reproduced to  $1.6 \times 10^{-14}$  ( $p = 0.44$  and  $0.32$ ), and the  $L = 6$  blocks against fresh unrestricted colour-restricted Arnoldi solves. The  $L = 8$  rung runs the block at the locator grid; the completed rung (5 grid points) is deposited as `v22_n5_L8_rung.json` and hashed in `certificate_sha256_v10.txt`.

Reproduction: run the scripts in this order (each writes its JSON to `results/v22-exactZ3-n5L8/` and its log to `logs/v22-exactZ3-n5L8/`):  
`v22_clifford_3design_v1.py`,  
`v22_z3_exact_v1.py` all, `v22_traj_beta2_v1.py`,  
`v22_n4_eps_topup_v1.py`, `v22_q4_logcorr_v1.py`,

`v22_tiltvar_uniform_v1.py`,  
`v22_n5_L8_block_v1.py` validate,  
`v22_n5_L8_block_v1.py` ids,  
`v22_n5_L8_block_v1.py` run --pgrid  
0.44,0.46,0.47,0.48,0.50. Dependencies: `numpy`,  
`scipy` (dense eigensolvers); the wall time is minutes per script except the  $L = 8$  block assembly (hours; the canonical-id pass is  $p$ -independent and done once). The files of this round are hashed in `certificate_sha256_v9.txt`; the  $v23$  rung completion (`v22_n5_L8_rung.json`, `manuscript_revised_v23_rung`) is hashed in `certificate_sha256_v10.txt`.

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- [1] S. Boucheron, G. Lugosi, and P. Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence* (Oxford University Press, Oxford, 2013).
  - [2] R. T. Rockafellar, *Convex Analysis* (Princeton University Press, Princeton, 1970).
  - [3] A. Dembo and O. Zeitouni, *Large Deviations Techniques and Applications*, 2nd ed. (Springer, New York, 1998).
  - [4] P. Del Moral, *Feynman–Kac Formulae: Genealogical and Interacting Particle Systems with Applications* (Springer, New York, 2004).

TABLE S8. Certificate files and SHA-256 hashes.

| file                                                    | SHA-256                                                           |
|---------------------------------------------------------|-------------------------------------------------------------------|
| n3_mult_cert_nb2.json                                   | 5600c82b82c7875210b9ab7803d0a5c8f6934d57f6afbc8f016679d6435ea56a  |
| n3_mult_cert_nb3.json                                   | 135635657992709144b16324a470977ed394a02bc5b21c11bc2da130294807de  |
| n3_mult_cert_nb4.json                                   | 2b618ab68941a68d53d6a0b678411f65053308c8eae8373a083aa55d155ad1dd  |
| n3_mult_cert_nb2_polys.json                             | 26fcc456ce8341f10d34c8c6abca08d04de9e0241984f7ccd43057d38e4a542a  |
| n3_mult_cert_nb3_polys.json                             | 016233445a4e8139c07dce862e1be358f8fa094df277517a942b14381b2eb837  |
| n3_mult_cert_nb4_polys.json                             | 9d944ec1dcacfb1a26216cae008fec698573cfa11de95935c084025abece505c  |
| n3_mult_cert_d3_nb2.json                                | 20cc93dfe011ab6fc20925a5310bdc3974f1aa5ab6853603c0e00e750cf89b30  |
| n3_mult_cert_d3_nb3.json                                | 7fcd62c254fff32a078dca08e268e16c254478918c6aa6c075d40228359bfe9   |
| n3_mult_cert_d3_nb4.json                                | f258be716d31e4a1d48472bfcdb364852eaeefcb8b0629733e82c4d84e7d0be78 |
| n3_mult_cert_d3_nb2_polys.json                          | 161e1ef5fc91debe27f598a6aee68199dbd35419fdc8fd4217204a5c7505973a  |
| n3_mult_cert_d3_nb3_polys.json                          | 7e23016ad43897e250fc6e089b02a84fc0eb36814d06184a8ed0c9f6ecc625d0  |
| n3_mult_cert_d3_nb4_polys.json                          | 1295e4deb729ce426eccad936240c71f7272a096c24344ccacbc6b81fd90c7    |
| n3_exceptional_exact_nb3.json                           | 55effc4a80ba193af63f449c6599bbf590e3bb0286bd015a79650b635c9996fa  |
| n3_exceptional_exact_nb4.json                           | 32442fe7d0dd851619d9a8b4c349c5595abb37df7c020b4594419f5657d66f22  |
| n3_exceptional_exact_d3_nb2.json                        | 11ce1f740cbe4d5907461c4bb730d4fd5f7e67b1d4a0a496d1c52738d119e9ee  |
| n3_exceptional_exact_d3_nb3.json                        | 785850b46a07fbde476eda6f26058fcee7e56c7ac34745a5f9e2488b2fe7390e  |
| n3_exceptional_exact_d3_nb4.json                        | e53211284b01a69f0f693296bff02d82fd5b4e42489f39012880100ab3bb8d1b  |
| n3_L12_kernel_verify_k3.json                            | 37c2581e753c771b51c84166f0decad50f9262028243d699ff7643a42b695666  |
| n3_L12_moving_kernel_2_Ind(triv.sgn)_minus.json         | 451e548b6b90b150419091f5e829d0e10a7ef9cee5f1abd56485c2386c9d98a1  |
| n3_L12_moving_kernel_4_Ind(sgn.std)_minus.json          | e685aa5e8e927203094af86049bbb093fac6f3200dd63a30046b824d2664555b  |
| n3_L12_moving_kernel_4_Ind(triv.std)_minus.json         | 9dd9a9d3cece96412693eefc9086858a03a2192402df6a442d7dd0a1a7f5321a  |
| n3_L12_moving_kernel_4_std.std_i-_minus.json            | 96c91b35b2fcb2905cfd0083d1217d12eba8900f09d979f1707c181943dbf55f  |
| n3_L12_moving_kernel_2_Ind(triv.sgn)_minus_d3.json      | 51cf134d4969294c65ffdaa941e2b6dc7769ef473060218a7f6ca783d9842280  |
| n3_L12_moving_kernel_4_Ind(sgn.std)_minus_d3.json       | 6eeb8cb6a8cb4f9a5734180e91eb0f720ee80a420a4ea9cdeb9e1b1f3c79abb7d |
| n3_L12_moving_kernel_4_Ind(triv.std)_minus_d3.json      | 32bc17daddb1981d9769e718825f6a92e7dae6c6c1963e71ffcac05ad0720e3d  |
| n3_L12_moving_kernel_4_std.std_i-_minus_d3.json         | afe7d826252cc60b536647ad15fc9cc059494540cdbe59686a374937168527a8  |
| n3_L12_kernel_1_sgn.sgn_i+_k3_q1048261_rational.json    | 0f49fed2443e23c4f446d46cdf30befa901ba16561561bc826d25d5f127e7e7b  |
| n3_L12_kernel_1_sgn.sgn_i-_k3_q1048261_rational.json    | 4acd6bc8d9bc19682a595e82403dc175f9912d86cd6cecc5b15334dfecfc981a  |
| n3_L12_kernel_1_triv.triv_i-_k3_q1048261_rational.json  | 1d2aa2f1901995a12c92e9acf19d0f71a87617f4a73dd7128ea7a09b35fec639  |
| n3_L12_kernel_2_Ind(triv.sgn)_k3_q1048261_rational.json | 9d02b012b4f1d9b631ec1bb305be7f7fea6c6c568a8ac12b295ba4926e939285  |
| n3_L12_kernel_4_Ind(sgn.std)_k3_q1048261_rational.json  | 5652eebdda79b9a665da7ba07e77385dadb55a88df8f272404a4065da18bc530  |
| n3_L12_kernel_4_Ind(triv.std)_k3_q1048261_rational.json | 82e14c6e6396ee75589fd265d422143f39cfccc3597d463bc0026d0f8ac1fa    |
| n3_L12_kernel_4_std.std_i+_k3_q1048261_rational.json    | 33be5db62064dbdc2088fff795f25b07ef3151060ea55e3d4f23a4f5fcb6b2e3  |
| n3_L12_kernel_4_std.std_i-_k3_q1048261_rational.json    | b5edc70bc1eb3f3d47c6b110b797d7ef84435f487c272fb9044e0ff4db9d4a45  |
| n3_L12_final.json                                       | 0aa3f5a6e678b8bc80b0cba4d0a84a170a54cbdb810c9d7e666a42068c795fd75 |
| n3_L12_final_d3.json                                    | 09dcd3fc785da31010975c85b7dcec2bafef6fd8cf62230882dd0ad06dae4aff  |
| n3_sharpness_L10.log                                    | b0969da19857824e0637ee1e1a634e45039d793019161eee1333f71397d0d81f  |
| n3_sharpness_L12.log                                    | f3c2f28467d4acf15ca952587bfbfb90ebf527b4d5e39e8299002fabd7ac0930  |
| n3_sharpness_L12_k3_central.log                         | 70664316b052b71c69e6ef4ac620e8f6309c65baea54e1232d68276ace414e8   |
| n3_sharpness_L12_k3_d3.log                              | 165d9079fb4a065396d0a49d9666143be90a6a36542c3230c0b95f15888868793 |
| n3_sharpness_L12_k3_p1-2.log                            | c28642ef7b473425a9bdaee11f53440890a083c15e85a10e9ed8d91d17ed3f11  |
| n3_sharpness_L12_k3_p1-3.log                            | 568ca64a32e15eb8007b2971fb4740eab400c2ec8d73776d25fb053a7bc1d8bf  |
| n3_sharpness_L12_k3_p3-5.log                            | 129d75f4cf9e8e9a9a49ff57ee704d9850f0fec87842a087bc1693bcbb6d1b85  |
| n3_sharpness_L12_d3_full.log                            | 36060cf646724100fd3bf8af19b28c7ba8a8dcb32057ec3bbc4d7332aafa3cf6  |
| n3_sharpness_L12_p0.log                                 | 53dfefbc3ccbf3a3dd0707b942640464528b497140e1bea339bcfd2c0dea1889  |
| n3_sharpness_L12_p0_q1048573.log                        | 1c7b9af12a9e6a3f9c4cfcb9070f73830c8dde1425dfff2e0377580f94e6bf6e  |
| n3_sharpness_L12_d3_px.log                              | 17fab566985c51be05fe068853813704e9e25fbb758ed95180cb96dd92e6aa0   |
| n3_sharpness_L12_d3_px_q1048573.log                     | 34532374c00dfa678be17c2ffe0c8cf9ae88ffd06b3585cbf307822e0efa865e  |
| n3_sharpness_L10_d3.log                                 | a1663286d77125fa8cf673084dce411df512aa9b3dd38e7fe9d4d976dbfe771d  |
| n3_sharpness_L10_d4.log                                 | af8c189121d9d15692f0add848aea65d63d96d27b52920e4ccca1c8902c9e2b7  |
| n3_sharpness_L10_d5.log                                 | a5999f67c25a9165dd946a9e5949fa185d17924e79100918e63a8530f7664234  |
| n3_sharpness_L10_d6.log                                 | 10292421bfc5316cd1fd28ca5f22d125f2fcc340773f5f7001bf03fd8e75a68b  |
| n3_sharpness_L10_d5_p1-3.log                            | cf435e2612e156df6600114189675f1853528bcfb4edf2b2281f95644f8b143e  |
| n3_sharpness_L4-10_d7_p1-3.log                          | 8b76837e6edfe453030ec5633a841661efed12a696f1d6fc36529873049096d6  |
| n3_sharpness_L4-8_d456_p1-3.log                         | 9f344ea8e8d7df8501d722fb23ecf0f3d02a2b637d17bbc2990d7850e084d18a  |

TABLE S9. Scripts and SHA-256 hashes.

| script                    | SHA-256                                                          |
|---------------------------|------------------------------------------------------------------|
| gap_utils.py              | 784e2aa0e2275c8fa9123073a1ac267b6ca71279231801a484b7cbf410e1a3d1 |
| n3_mult_certificates.py   | ed2abc22e9dbcc091372e1845b8a9ca2fcc0cd271ec4f9c01040a37f77b9b561 |
| n3_mult_certificates_d.py | 81e8b91b0de3df0140edbc21848ccd7d42923c0b76331cc182b7e5b7f51825c  |
| n3_exceptional_exact.py   | 7b82757b018b9ac74ea963f66091eda3034c363aef842a0e74a1fd1fceb5cadd |
| n3_exceptional_exact_d.py | 1753756feddbce19e4f630e933d0fcbe169e3a2c46776be41ac73a0a5677a791 |
| n3_simple_zero_check.py   | 30fc0dfa4795e5e637b252516f48b95c0f5f9e7dbcd4b703ffab2fd9d9cb35   |
| n3_p0_exact_rank.py       | adedb74bf3b3956cd4d2b77392950475a2c80da7bb449f776b3bf69c0e8ba582 |
| n3_closed_form_proof.py   | da8cc292493b9d7bc030b13a31a81b8289fd5e940beca19900b05e4abb3eb5d2 |
| n3_formula_exact.py       | 2accd8a1e5360fcba4982138b4973ed2f4848cb1cb7b270f91d85dacc144dfe3 |
| n3_sharpness_L12.py       | 3dffec312836e13489c48889d3209a862c207314f2fbc3ff6f63cebd8cf350d3 |
| n3_L12_structure_check.py | bd5064ad249d9589f5200ec9e4c5203392176391f46014f3aec46a15d8e7d312 |
| n3_L12_kernel.py          | dd962b398a3888f1c6b1c26368c91a9af59c63a5d9ebc3a302f4f0e53ae581e5 |
| n3_L12_kernel_verify.py   | 548dd40a29b15c7d0fd8188f1e750aeb82fb210006054bcfed40f6fa949cad0f |
| n3_L12_moving_kernel.py   | 07b13b7a3eac81d647290a15c86fd122bcacdc2db7ba40bea868f193550800be |
| n3_L12_final.py           | 613584dd468066855a842c6ddcf42535bb8d428bfd9de18659422d0f742e9087 |
| n3_kernel_recheck.py      | 30062ba34504253b2d770d5bfb795d38371ad1aeab1bafc92fd8a467c142079  |

TABLE S10. Calibration of the exact evolution and of the trajectory simulator against the deposited tilt chain (script `mipr_scgf_exact.py`, run logs `mipr_scgf_exact.log`, `mipr_born_scgf.log`). The deposited exact values are reproduced to their rounding at both sizes.

| quantity                                                    | $L = 8, t = 4$            | deposit                 | $L = 12, t = 6$          | deposit  |
|-------------------------------------------------------------|---------------------------|-------------------------|--------------------------|----------|
| $Z_2$ (exact evolution)                                     | $1.479639 \times 10^{-2}$ | $1.4796 \times 10^{-2}$ | $3.81731 \times 10^{-5}$ | —        |
| $\tilde{S}_2$ (exact)                                       | 2.1547                    | 2.155                   | 3.1281                   | 3.128    |
| $\mathbb{E}[2^{-N_{\text{rand}}}]$ ( $4 \times 10^4$ traj.) | $1.4689 \times 10^{-2}$   | $1.4942 \times 10^{-2}$ | $3.8399 \times 10^{-5}$  | —        |
| $\mathbb{E}S_{L/2}$ (traj.)                                 | 2.015(4)                  | 2.019(4)                | 2.632(5)                 | 2.394(5) |
| ESS (tilt)                                                  | 6339                      | 5935                    | 478                      | 454      |

TABLE S11. Exact  $\beta = 1$  anchors of  $\Lambda_L$  (`scgf_exact.json`, `scgf_exact_addendum.json`): finite- $t$  values at  $t = \tau L$  from the  $3^L$  evolution ( $L \leq 16$ ), the asymptotic  $\log \lambda_1(L)/(2L)$ , and the tail amplitude  $A(L) = \tilde{Z}_2(t)\lambda_1^{-t}$  (constant to six digits for  $t \gtrsim 3L$ ; `z2_tail` in `scgf_exact.json`).

| $L$ | $p$  | $t$ | $2Lt \Lambda_L(1)$ fin. $t$ | $\log \lambda_1/(2L)$ | $A(L)$           |
|-----|------|-----|-----------------------------|-----------------------|------------------|
| 8   | 0.16 | 32  | -0.077984                   | -0.079735             | 2.4512           |
| 12  | 0.16 | 48  | -0.079362                   | -0.080615             | 4.2354           |
| 16  | 0.16 | 64  | -0.079901                   | -0.080867             | 7.2288           |
| 24  | 0.16 | 96  | -0.080325 <sup>a</sup>      | -0.080989             | $\approx 21.4^b$ |
| 8   | 0.22 | 32  | -0.106382                   | -0.108701             | 3.2786           |
| 12  | 0.22 | 48  | -0.108278                   | -0.109947             | 6.8326           |

<sup>a</sup>Amplitude-corrected  $\lambda_1$  anchor,  $\log \lambda_1/(2L) + \log A/(2Lt)$ .

<sup>b</sup>Extrapolated from  $\log A(L) \approx 0.135 L$   
( $\log A = 0.896, 1.443, 1.979$  at  $L = 8, 12, 16$ ).

TABLE S12. The production ladder (`scgf_born.json`;  $t = 4L$ ;  $\bar{x}_L$  and  $\text{Var}/(2Lt)$  in bits per site and bits<sup>2</sup> per site;  $\Lambda_L$  values in nats per site; bootstrap s.e. (200 resamples) on  $\bar{x}_L$  at the  $10^{-3}$  level).  $g_L$  uses the exact finite- $t$  anchors of Table S11 ( $L \leq 16$ ) and the amplitude-corrected value at  $L = 24$ . The ESS column is the within-sample collision-tilt effective sample size at  $\beta = 1$ .

| $L$ | $p$  | $B$   | $\bar{x}_L$ | $\text{Var}/(2Lt)$ | $\Lambda_L(0.25)$ | $\Lambda_L(0.5)$ | $\Lambda_L(1)$        | $g_L$   | $\text{ESS}_{\beta=1}$ |
|-----|------|-------|-------------|--------------------|-------------------|------------------|-----------------------|---------|------------------------|
| 8   | 0.16 | 40000 | 0.14479     | 0.10547            | -0.02357          | -0.04440         | -0.07798              | 0.02238 | 11.7                   |
| 12  | 0.16 | 40000 | 0.14703     | 0.10441            | -0.02392          | -0.04495         | -0.07936              | 0.02255 | 3.9                    |
| 16  | 0.16 | 30000 | 0.14803     | 0.10581            | -0.02412          | -0.04586         | -0.07990              | 0.02270 | 1.4                    |
| 24  | 0.16 | 15000 | 0.14890     | 0.10481            | -0.02428          | -0.04689         | -0.08033 <sup>a</sup> | 0.02288 | 1.0                    |
| 8   | 0.22 | 40000 | 0.19280     | 0.12532            | -0.03158          | -0.05988         | -0.10638              | 0.02726 | 18.2                   |
| 12  | 0.22 | 40000 | 0.19512     | 0.12415            | -0.03203          | -0.06110         | -0.10828              | 0.02697 | 2.7                    |
| 16  | 0.22 | 30000 | 0.19614     | 0.12409            | -0.03219          | -0.06165         | —                     | —       | 1.1                    |
| 24  | 0.22 | 15000 | 0.19706     | 0.12365            | -0.03225          | -0.06274         | —                     | —       | 1.0                    |